



# NSMA - An efficient hybrid NSGA-II and Local search algorithm for a multi-objective Hub Location Problem

Optimizing Thailand railway transport infrastructure, for the new high-speed rail (HSR) line along the Northern Line

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# Introduction

- Hub and Spoke network are widely used for transport, telecom and computers
- With *economies of scales*, costs are saved by transferring consolidated flows
- Hub Location Problem (HLP) belongs to the **NP**-hard class
- A *multi-objective* HLP is used to model the **railway** system
- New meta-heuristics algorithm is proposed to provide optimal trade-off solutions

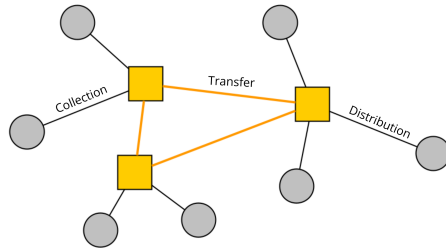


Figure: Hub and Spoke network

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## Related Works

- O'Kelly [9] proposed the first formulation for HLP. In 1994, Campbell [3] introduced *p-hub center* and hub covering problem. For more insights, please refer to the survey [2].
- In reality, it is more beneficial to build **incomplete** hub network. Alumur [1] gives formulations for the single allocation hub location problems.
- To solve real-world transportation problems, many authors developed different **multi-objective** models and solution methods. Demir [5] proposes a bi-objective multiple allocation hub network design and routing problem and used NSGA-II algorithm for solution. Chanta [10] presents a biobjective hub location model that considers congestion for **railway** transportation planning.
  - ▷ In this work, we formulate a tri-objective single allocation hub location and network design problem, with service-based objectives considering congestion.
  - ▷ Moreover, we propose a hybrid algorithm **NSMA** that combines NSGA-II and local search to solve the problem efficiently.

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## Input parameters

- $N$  : set of nodes, indexed by  $1, 2, 3, \dots, n$
- $H \subseteq N$  : set of potential hub nodes, indexed  $1, 2, \dots, h$
- $w$  : demand (flow) matrix,  $w_{ij}$  is the amount of flow to be sent from origin node  $i \in N$  to destination node  $j \in N$ . For convenience, define:
  - $O_i = \sum_{j \in N} w_{ij}$  : flow originating from node  $i$
  - $D_i = \sum_{j \in N} w_{ji}$  : flow destined for node  $i$
- $C_{ij}, t_{ij}$  : transportation cost, travel time between  $i \in N, j \in N$ , respectively
- $\alpha, \beta, \chi, \delta$  : discount factor for transportation cost on hub links, discount factor for traveling time on hub link, collection coefficient, distribution coefficient
- $FH_k$  fixed cost for opening a hub with unlimited capacity at node  $k$  ( $k \in H$ )
- $FL_{ij}$  fixed cost to open a link between hub  $i \in H$  and  $j \in H$
- $t_k$  : the delay time at hub node  $k \in H$  [10]
- $L$  : constant distance defining demand coverage

# Assumptions

- ① Each node is allocated to exactly one hub (single allocation)
- ② The hubs network are *strongly* connected, i.e., there exists a directed path from any hub node  $i \in H$  to every other hub node  $j \in H$  via the hub network
  - Hub arcs are uni-directional
  - Cost matrix  $C$  does not guarantee symmetry
- ③ Direct connection between non-hub nodes is disallowed
- ④ Cost matrix  $C$  satisfies triangle inequality, i.e.  $C_{ij} \leq C_{ik} + C_{kj} \quad \forall i, j, k \in N$



## Decision variables

Denote:

- $x_{ij} = 1$  if node  $i \in N$  is allocated to the hub node  $j \in H$ , 0 otherwise  
 $x_{kk} = 1$  if node  $k$  is a hub.
- $z_{ij} = 1$  if the hub arc from  $i$  to  $j$  is established ( $i, j \in H$ ), 0 otherwise;
- $f_{ij}^k$  the total amount of flow (*commodity*) originating from node  $k \in N$  and to be routed on hub arc  $(i, j)$  in that order (from  $i$  to  $j$ )

To formulate the p-hub center and covering problem on incomplete hub network, Alumur [1] defined the following variables:

- $y_{ij}^k = 1$  if the spanning tree rooted at hub  $k \in H$  uses the hub arc  $(i, j)$ ; else 0
- $s_{ij}$  the undiscounted travel time from hub  $i \in H$  to hub  $j \in H$  in the hub network

For convenience, let us denote  $X_i = \sum_{k \in H} kx_{ik}$  index of the hub node allocated to  $i$ .  $X$  concides with the *allocation vector*, which we will see in Section 4

# Objectives

$$\text{Minimize } Z_1 = \sum_{i \in H} \sum_{j \in H} FL_{ij} z_{ij} + \sum_{k \in H} FH_k x_{kk} + \sum_{i \in N} (\chi C_{ik} O_i + \delta C_{ki} D_i) x_i x_i + \sum_{i \in H} \sum_{j \in H} \sum_{k \in N} \alpha C_{ij} f_{ij}^k$$

$$\text{Minimize } Z_2 = \max_{i,j \in N} \{t_i x_i + \beta s_{x_i x_j} + t_{x_j j}\}$$

$$\text{Maximize } Z_3 = \sum_{i \in N} O_i [C_i x_i \leq L]$$

(note:  $[B] = 1$  if  $B$  is true, 0 otherwise. It is a binary variable)

$Z_1$  is the total cost, including fixed costs and transportation costs.

$Z_2$  is the maximum travel time of an O-D pair  $(i, j)$ .

$Z_3$  is the amount of demand covered by the system.

# Constraints

Subjects to:

$$\sum_{j \in H} x_{ij} = 1 \quad \forall i \in N \quad (1)$$

$$x_{ik} \leq x_{kk} \quad \forall i \in N, k \in H \quad (2)$$

$$z_{ij} \leq x_{ii} + x_{jj} - 1 \quad \forall i, j \in H \quad (3)$$

$$\sum_{j \in H: j \neq i} f_{ji}^k + O_k x_{ki} = \sum_{j \in H: j \neq i} f_{ij}^k + \sum_{l \in N} w_{kl} x_{li} \quad \forall i \in H, k \in N \quad (4)$$

$$f_{ij}^k \leq O_k z_{ij} \quad \forall i, j \in H, k \in N \quad (5)$$

Equation (1) ensures single allocation property. Eq. (2), (3) ensures valid allocation and establishing of arcs. Eq. (4) is the flow divergence equation, and Eq. (5) ensures flow can only pass through established arcs.

## Constraints (adapted from [1])

$$\sum_{i \in H, i \neq j} y_{ij}^k \geq x_{jj} + x_{kk} - 1 \quad \forall j, k \in H : j \neq k \quad (6)$$

$$\sum_{i \in H, i \neq j} y_{ij}^k \leq x_{kk} \quad \forall j, k \in H : j \neq k \quad (7)$$

$$y_{ij}^k \leq z_{ij} \quad \forall i, j, k \in H \quad (8)$$

$$y_{ij}^k + y_{ji}^k \leq 1 \quad \forall i, j, k \in H \quad (9)$$

$$s_{kj} \geq (s_{ki} + t_i + t_{ij})y_{ij}^k \quad \forall i, j, k \in H : i \neq j \wedge j \neq k \quad (10)$$

$$s_{kk} = 0 \quad \forall k \in H \quad (11)$$

The constraints (6) – (11) constructs a directed spanning tree rooted at each hub  $k$  based on  $z_{ij}$ . Then, shortest distances are computed via  $s$  using edge-relaxation (10).

## Characteristics of the model

- The mathematical formulation has  $2n^3 + 3n^2$  variables,  $n^3 + 3n^2$  of which are binary. It has  $O(n^3)$  constraints
- Under single objective, the given formulation resembles *Mixed Integer Programming* model.
- Using  $\epsilon$ -constraint method, one can solve the above multi-objective model using some commercial solver software tools.
- This model is an adaptation of the work by Alumur [1]
- Consideration for congestion (delay time  $t_i$  [10] at hub nodes) are embedded in (10)

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# Memetic algorithm

Memetic Algorithm = Genetic Algorithm + Local Search

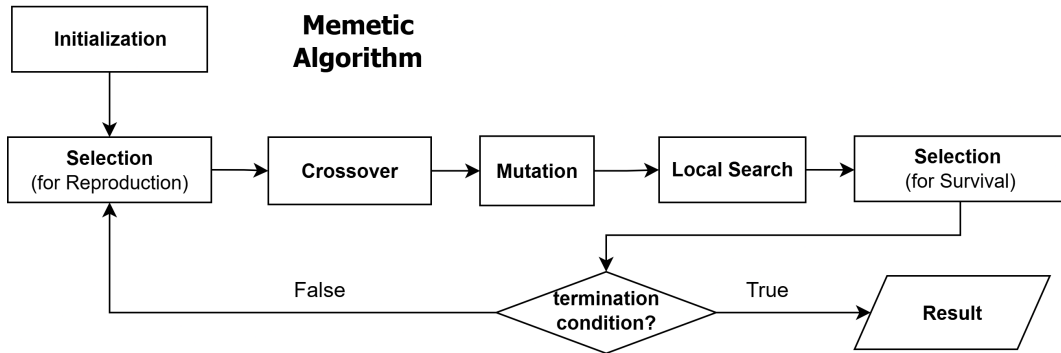


Figure: Steps in a Memetic Algorithm

# NSGA-II

Properties of NSGA-II [4]:

- One of the most well-known multi-objective evolutionary algorithms (MOEAs)
- **Elitist:** NSGA-II combines parents and offspring, selecting the best for survival.
- It uses fast non-dominated sorting (in  $O(MN^2)$  instead of traditional  $O(MN^3)$ ) to generate non-domination layers  $F_1, F_2, F_3, \dots$
- Introduced crowding distance to promote diversity,  $i_{distance} = \sum_{m=1}^M \frac{x_{high}^m - x_{low}^m}{f_{max}^m - f_{min}^m}$ , where  $x_{low}, x_{high}$  is the two points nearest to solution  $i$  in objective space  $m$ .  
 $i_{distance} = \infty$  if one of its objective value is extreme.
- Ranking and distance are used in selection phases of algorithm.  $i \prec_n j$  if  $i_{rank} < j_{rank}$  or  $i_{rank} = j_{rank}$  and  $i_{distance} > j_{distance}$

The NSGA-II algorithm uses binary tournament selection (based on  $\prec_n$ ) and the proposed elitist selection process. The crossover and mutation operators need to be *designed* for specific application.



# Representation of solution - Part I: Hub location

This work uses *allocation vector* representation.

- $A_i = k$  ( $k \neq i$ ) if node  $i$  is allocated to hub  $k$ ; otherwise,  $A_k = k$  if  $k$  is a hub
- The representation must satisfy constraint 2 (i.e.  $A_{A_i} = A_i$ )

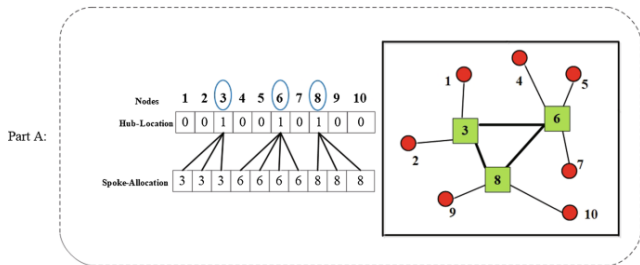


Figure: Allocation vector representation, image source: [12]

## Solution representation - Part II : hub network (arc location)

Representation:

- Denote  $\mathcal{H}(A) = \{h_1, h_2, \dots, h_p\}$  the hubs in the representation  $A$ , then  $p = |\mathcal{H}|$  is the number of hubs.
- We use a binary string of length  $p(p-1)$  bits to represent the state of each arc  $(h_i, h_j)$ . If  $(h_i, h_j)$  are in the network, then the corresponding bit is 1, otherwise 0.

Evaluation of objective values:

We assume the representation is valid, if not, apply the `repair()` method

- Compute all-pair shortest paths using Floyd algorithm [6] in  $O(n^3)$
- Loop over every O-D pair  $(i, j)$  and compute  $Z_1, Z_2, Z_3$  in  $O(n^2)$
- Total time complexity for evaluation is  $O(n^3)$

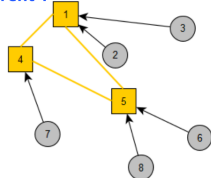
## Crossover operator - Phase I (Hub Crossover)

The steps of the crossover operator on parents  $(P_1, P_2)$  to create children  $(C_1, C_2)$  is as follows:

- 1 **Hub exchange:** randomly distribute  $\mathcal{H}(P_1) \cup \mathcal{H}(P_2)$  to  $\mathcal{H}(C_1)$  and  $\mathcal{H}(C_2)$  with uniform probability. The common hubs  $\mathcal{H}(P_1) \cap \mathcal{H}(P_2)$  are inherited in both  $C_1, C_2$
- 2 **Hub mapping:** create mappings  $\mathcal{M}_{C_1}, \mathcal{M}_{C_2}$  for  $C_1, C_2$  respectively.  $\mathcal{M}_A(k) = k$  if  $k \in \mathcal{H}(A)$ , else  $\mathcal{M}_A(h) = \arg \min_{k \in \mathcal{H}(A)} \text{distance}(k, h)$ . If a hub  $h$  is not present in  $\mathcal{H}$ , then it is mapped to the nearest hub  $k \in \mathcal{H}$ .
- 3 **Crossover:**  $C_1 \leftarrow P_1; C_2 \leftarrow P_2$ , then 1-point crossover is applied on the allocation vectors  $C_1, C_2$ .
- 4 **Repair:**  $C_1$  and  $C_2$ 's allocations are fixed according to  $\mathcal{M}_{C_1}$  and  $\mathcal{M}_{C_2}$ , ensuring nodes are assigned to proper hub.

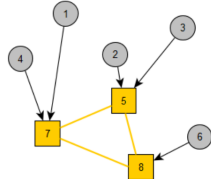
# Example - phase I of crossover operator

Parent 1

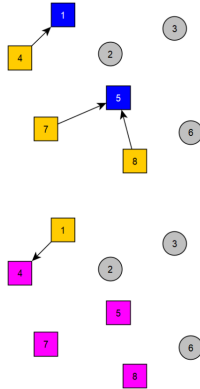


|    |   |   |   |   |   |   |   |   |
|----|---|---|---|---|---|---|---|---|
|    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| p1 | 1 | 1 | 1 | 4 | 5 | 5 | 4 | 5 |
|    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| p2 | 7 | 5 | 5 | 7 | 5 | 8 | 7 | 8 |

Parent 2



Hub exchange & mapping



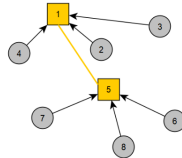
Allocation crossover

|    |   |   |   |   |   |   |   |   |
|----|---|---|---|---|---|---|---|---|
|    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| c1 | 1 | 1 | 1 | 4 | 5 | 8 | 7 | 8 |
| c2 | 7 | 5 | 5 | 7 | 5 | 5 | 4 | 5 |

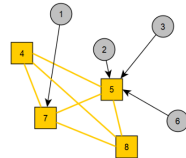
repair

|    |   |   |   |   |   |   |   |   |
|----|---|---|---|---|---|---|---|---|
|    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| c1 | 1 | 1 | 1 | 1 | 5 | 5 | 5 | 5 |
| c2 | 7 | 5 | 5 | 4 | 5 | 5 | 7 | 8 |

Child 1



Child 2



## Crossover - Phase II (Arcs Crossover)

Denote  $\mathcal{A}(A)$  the arcs in the representation  $A$ . In a similar manner, the second part of representation is crossed over as described below:

- 1 **Arc exchange:** randomly distribute  $\mathcal{A}(P_1) \cup \mathcal{A}(P_2)$  to  $\mathcal{A}(C_1)$  and  $\mathcal{A}(C_2)$  with ratio  $|\mathcal{H}(C_1)|^2/|\mathcal{H}(C_2)|^2$ . The common arcs  $\mathcal{A}(P_1) \cap \mathcal{A}(P_2)$  are inherited in both  $C_1, C_2$ .
- 2 **Arc mapping:** each arc  $(u, v)$  is mapped to  $(\mathcal{M}_C(u), \mathcal{M}_C(v))$ . Invalid self-loops and multiple edges are removed.
- 3 **Repair:** by fixing the strong connectivity property (the `repair()` method).

# Mutation operators

The four mutation operators below has the same probability to be chosen ( $\frac{1}{4}$ ):

- **m1 - Make hub**: select a random node  $i$  and make it a hub
- **m2 - Erase hub**: select a random hub  $k$  and make it a node. All node  $j$  currently allocated to  $k$  is then assigned to a random hub (other than  $k$ ).
- **m3 - Hub exchange**: exchange a hub  $k$  with a non-hub node  $i$ . All nodes currently assigned to  $k$  will be allocated to  $i$ .
- **m4 - Arc flip**: selects  $\max\{2, \frac{|A|}{p}\}$  random bits in the arc representation and flips them. If the flipped bit is 1, then the corresponding arc is removed, otherwise it is added.

# Mutation operators illustration

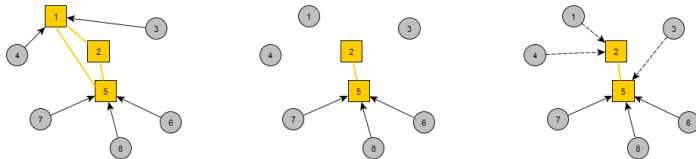


Figure: m2 erases hub 1

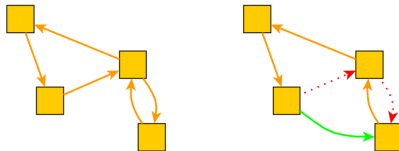


Figure: m4 flips 3 bits, adding 1 arcs and remove 2 arcs

## Mutation operators illustration (cont'd)

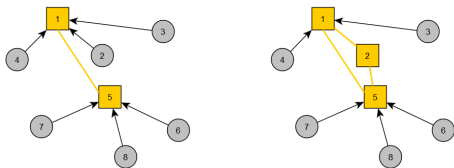


Figure: m1 makes new hub 2

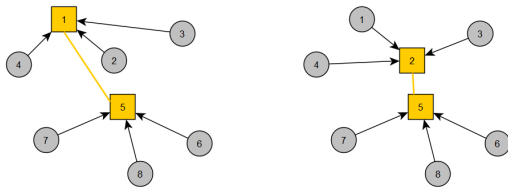


Figure: m3 exchanges hub 1 with node 2



## Repair operator - Fix connectivity

Crossover and mutation (m4) operators may break the strong connectivity property of the hub network. To fix this, we apply the `repair()` method:

### `repair()` method - fix strong connectivity

Denote  $\bar{\mathcal{A}}$  the set of arcs **not** included in the representation  $A$ ;  
and  $a \rightsquigarrow b$  if node  $b$  is reachable from node  $a$  via the arcs  $\mathcal{A}$

```
1: for  $k \in H : A_k = k$  do                                ▷ for each hub  $k$  established
2:   while not  $k \rightsquigarrow \mathcal{H} \setminus \{k\}$  do                ▷ there is node un-reachable from  $k$ 
3:      $(u, v) \leftarrow \arg \min_{i, j \in \mathcal{H}} \{C_{ij} \mid (i, j) \in \bar{\mathcal{A}} \mid k \rightsquigarrow i \wedge \neg k \rightsquigarrow j\}$     ▷ min-cost arc
4:     Add the arc  $(u, v)$  to the representation  $A$ 
5:      $\bar{\mathcal{A}} \leftarrow \bar{\mathcal{A}} \setminus \{(u, v)\}$ 
6:     Perform Depth-first Search to update reachability status
7:   end while
8: end for
```

# NSGA-II algorithm for HLP

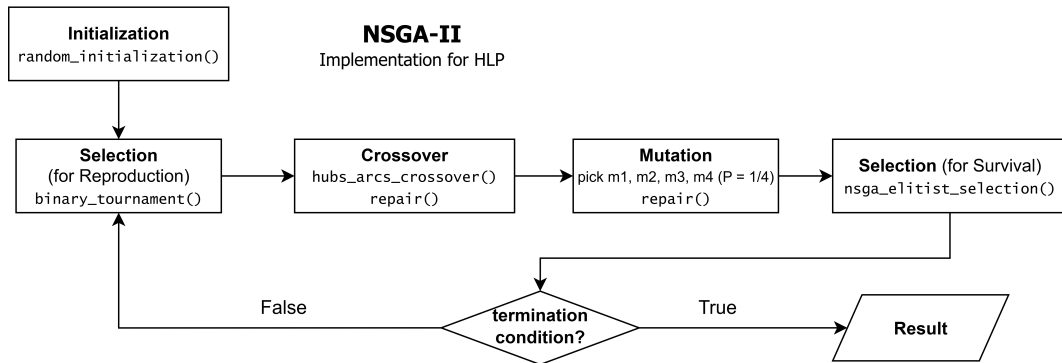


Figure: Detailed flow of NSGA-II algorithm for HLP

# Local search - Neighborhood structure

## Problem

- The proposed crossover operator can preserve traits from both parents
- Mutation makes small changes to the hub location and network structure
- However, allocation pattern is not optimized, leading to sub-optimal solutions

▷ The solution is Local Search!

Neighborhood of  $A$  is the set of representation differs from  $A$  at exactly 1 position.

$$\mathcal{N}(A) = \{A' \mid \text{distance}(A, A') = 1\}$$

## Apply local search in NSMA

$$\mathcal{N}(A) = \{A' \mid \text{distance}(A, A') = 1\}$$

- A representation  $A$  has exactly  $(n - p)(p - 1)$  neighbors ( $p$  is the number of hubs)
- To optimize the allocation vector given a fixed hubs network, loops through all neighbors iteratively and select the *first* one that *dominates* current solution.
- Repeat the process until no more changes is found
- Local search is applied after every mutation and for every 25 generations
- **Problem:** However, for each neighbor, simply invoking `evaluate()` method will incur a time complexity of  $O(n^3)$

# Efficient local search implementation

## Observation

Changing allocation  $A_i$  of node  $i$  from  $k$  to  $l$  only affect  $O(n)$  components

- $C_i A_i (\chi O_i + \delta D_i) + \sum_j \alpha s C_{A_i A_j} w_{ij} (Z_1)$
- $\{t_i A_i + \beta s_{A_i A_j} + t_{A_j j}\} (Z_2)$
- $O_i [C_i A_i \leq L] (Z_3)$

The implementation is then as follows:

- Compute changes to  $Z_1, Z_3$  by subtracting and adding corresponding components
- For  $Z_2$ , maintain  $\mathcal{T}_i = \{tt_{ij}\}$  - set of travel time to  $j$  for each  $i$ . Removes the  $i$ -th row and  $i$ -th column of the matrix  $\mathcal{T}$ , then compute max of remaining element, let it  $M = \max_{j \neq i, k \neq i} t_{ij}$ . For each new value of  $A_i$ , computes  $\mathcal{T}'_i$  then  $Z_2 = \max\{M, \max \mathcal{T}'_i\}$ . To compute  $M$  fast, use Segment Tree

Evaluation cost for each neighbor is now  $O(n)$  !

## Survival - removing duplicates

As a natural phenomenon, genetic algorithms will converge and eventually generates duplicate solutions in the population

- To avoid this and overcome *premature convergence*, we define the following `remove_duplicates()` procedure.

The `remove_duplicates()` method:

- ① At start of each generation, increase age of every individual by 1
- ② At the end of each generation, each duplicate solutions has a probability of  $\frac{\text{age}}{50}$  to be removed from the population
- ③ Thus, each duplicate solution has a life expectancy of  $\approx 8.543$  generations.
- ④ If number of solutions drops below `POP_SIZE`, random solutions are introduced to the population

# Initialization

It is well-accepted that initial population plays an important role in the optimization process.

The initial population is generated by the following steps:

- ① Generate a pool of size  $10 \times \text{POP\_SIZE}$
- ② Apply local search on each generated solution with a probability of 0.1.
- ③ Apply `nsga_elitist_selection()` method to select `POP_SIZE` best solution

After this process, we obtain a high-quality and diverse initial set of solutions.

# NSMA algorithm's detailed flow

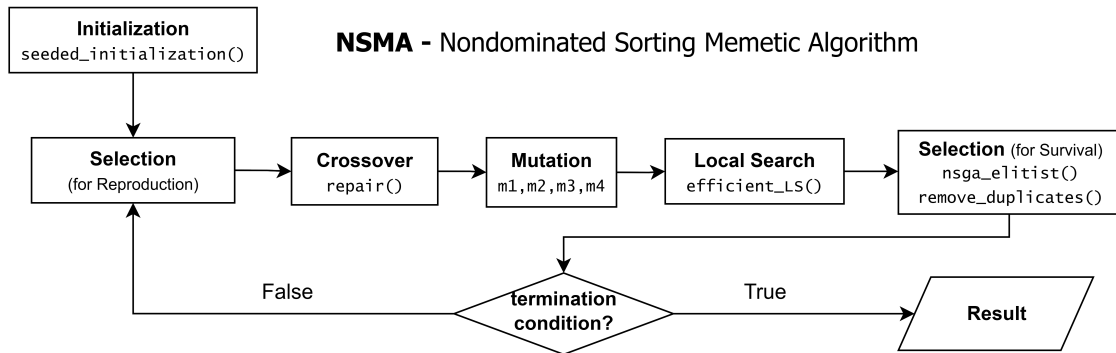


Figure: Detailed steps of NSMA algorithm



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## Dataset for experiment

- The dataset is taken from [10], consists of 54 train stations along The Northern Line. The distance matrix is calculated using Haversine formula. The flow matrix contains the averaged amount of passenger from each origin station  $i$  to destination station  $j$  over a period of years.
- From this dataset, two problem instances with size  $n = 10$  and  $n = 54$  are derived for each variant. The parameter FL is only used in the Incomplete variant.

| $n$ | $L$ | $FH_k$ | $FL_{ij}$            | Hub capacity | $t_k$ |
|-----|-----|--------|----------------------|--------------|-------|
| 10  | 30  | $10^6$ | $10^5 \times C_{ij}$ | Unlimited    | 10    |
| 54  | 60  | $10^7$ | $10^6 \times C_{ij}$ | Unlimited    | 10    |

Table: Input parameters for instances

## Experiment settings

- To have statistical significance and reliable measurement, each algorithm is run for 20 times on each instance
- Experiments are run on a laptop with CPU 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz with 16 GB RAM
- Algorithm parameters:

| Parameter            | NSGA-II                                              | NSMA |
|----------------------|------------------------------------------------------|------|
| POP_SIZE             | 500                                                  | 500  |
| NUM_GEN              | 200                                                  | 200  |
| $p_c$                | 0.95                                                 | 0.95 |
| $p_m$                | 0.3                                                  | 0.24 |
| $[f_{min}, f_{max}]$ | $\{[5 \cdot 10^8, 10^{10}], [20, 800], [0, 93000]\}$ |      |

Table: Algorithm parameters

# Evaluation metrics

- ① Size of Pareto set  
*larger is better*  
Larger set indicates more diverse solutions
- ② Hyper Volume indicator (HV)  
*larger is better*  
HV indicates the objective space dominated by the points set
- ③ Generational Distance (GD)  
*smaller is better*  
how well the algorithm approximates to the references?
- ④ Inverted Generational Distance (IGD)  
*smaller is better*  
how well the reference set being approximated?
- ⑤ Spacing  
*smaller is better*  
measures the uniformity of distribution of points  
computed by taking the standard deviation of the nearest-neighbor distances

# Hyper Volume indicator

## Definition (given in [7])

Given a point set  $S \subset \mathbb{R}^d$  and a reference point  $r \in \mathbb{R}^d$ , the hypervolume indicator of  $S$  is the measure of the region weakly dominated by  $S$  and bounded above by  $r$ , i.e.

$$HV(S) = \Lambda(\{q \in \mathbb{R}^d \mid \exists p \in S : p \leq q \text{ and } q \leq r\})$$

Where  $\Lambda$  denotes the Lebesgue measure. Alternatively (as union of boxes):

$$HV(S) = \Lambda \left( \bigcup_{p \in S \wedge p \leq r} [p, r] \right)$$

where  $[p, r]$  is the box delimited below by  $p \in S$  and above by  $r$

# Generational Distance (GD)

## Definition [8]

Denote  $R = \{r_1, r_2, \dots, r_{|R|}\}$  the reference point set (Pareto-optimal points); and  $A = \{a_1, a_2, \dots, a_{|A|}\}$  the set of non-dominated points found by our algorithm.

$$GD(A, R) = \frac{1}{|A|} \left( \sum_{i=1}^{|A|} d_i^p \right)^{1/p}$$

With  $p = 2$ ,  $d_i$  is the Euclidian distance from  $a_i$  to its nearest point in the reference set.  $d_i = \min_{j=1}^{|R|} \|a_i - r_j\|_2$

▷ Given the same  $R$ ,  $GD(A, R) < GD(B, R)$  means that solution set  $A$  has better convergence to  $R$ , compared to set  $B$

# Inverted Generational Distance (IGD)

## Definition [8]

IGD inverts the distances in GD and measures the average distance from any point in  $R$  to the closest point in  $A$ .

$$\text{IGD}(A, R) = \frac{1}{|R|} \left( \sum_{i=1}^{|R|} \hat{d}_i^p \right)^{1/p}$$

in which  $\hat{d}_i$  is distance from  $r_i$  to its closest point in  $A$ :  $\hat{d}_i = \min_{j=1}^{|A|} \|r_i - a_j\|_2$

- Besides convergence, IGD also accounts for coverage of reference points. *“If no solution associated with a reference point is found, the IGD metric value for the set will be large”*
- Smaller value of IGD is better

## Characteristics of metrics

| Metrics | Requires reference set $R$ | Requires reference point $r$ | Goal     | Time complexity             |
|---------|----------------------------|------------------------------|----------|-----------------------------|
| Size    | No                         | No                           | Maximize | $\mathcal{O}(1)$            |
| HV      | No                         | Yes                          | Maximize | $\mathcal{O}(n^{d-1})$ [11] |
| GD      | Yes                        | No                           | Minimize | $\mathcal{O}(n^2)$          |
| IGD     | Yes                        | No                           | Minimize | $\mathcal{O}(n^2)$          |
| Spacing | No                         | No                           | Minimize | $\mathcal{O}(n^2)$          |

Table: An overview of used evaluation metrics



## Settings for computation of metrics

- The true Pareto-front is unknown before-hand for large, complex real-world problem instances
- For each instance (each value of  $n$ ), the reference set  $R_n$  is approximated by taking the non-dominated (best) solutions in the combined sets of solutions found by both algorithms in all 20 runs.
- To handle different objective value scales, we used min-max normalization to convert all coordinates to range  $[0, 1]$  using formula:

$$x' = \frac{x - \min}{\max - \min}$$

## Results

Result for Complete Hub network (Uncapacitated Single Allocation HLP)

| n  | size |      | HV    |       | Spacing ( $10^{-2}$ ) |      | GD ( $10^{-3}$ ) |       | IGD ( $10^{-4}$ ) |      |
|----|------|------|-------|-------|-----------------------|------|------------------|-------|-------------------|------|
|    | NSGA | NSMA | NSGA  | NSMA  | NSGA                  | NSMA | NSGA             | NSMA  | NSGA              | NSMA |
| 10 | 17.4 | 19.1 | 0.801 | 0.804 | 16                    | 16   | 1.41             | 0.805 | 156               | 109  |
| 54 | 271  | 343  | 0.927 | 0.931 | 2.45                  | 2.04 | 1.64             | 0.729 | 9.94              | 7.91 |

Result for Incomplete Hub network variant

| n  | size |      | HV    |       | Spacing ( $10^{-2}$ ) |       | GD ( $10^{-3}$ ) |       | IGD ( $10^{-4}$ ) |      |
|----|------|------|-------|-------|-----------------------|-------|------------------|-------|-------------------|------|
|    | NSGA | NSMA | NSGA  | NSMA  | NSGA                  | NSMA  | NSGA             | NSMA  | NSGA              | NSMA |
| 10 | 5.5  | 7.55 | 0     | 0     | 0.159                 | 0.154 | 7.75             | 12.4  | 682               | 389  |
| 54 | 431  | 464  | 0.945 | 0.955 | 2.07                  | 1.68  | 1.16             | 0.452 | 7.35              | 5.39 |

Table: Evaluation metrics of NSGA-II and NSMA on two problem variants

## Results analysis

- NSMA's metrics have lower standard deviations than NSGA's
- NSMA has higher HV than NSGA-II on average
- NSMA generates more non-dominated points, distribute them evenly and has better convergence towards *optimal* Pareto front
- The result demonstrated the efficiency of Local Search in improving searching capability of the algorithm.

# Pareto front visualization

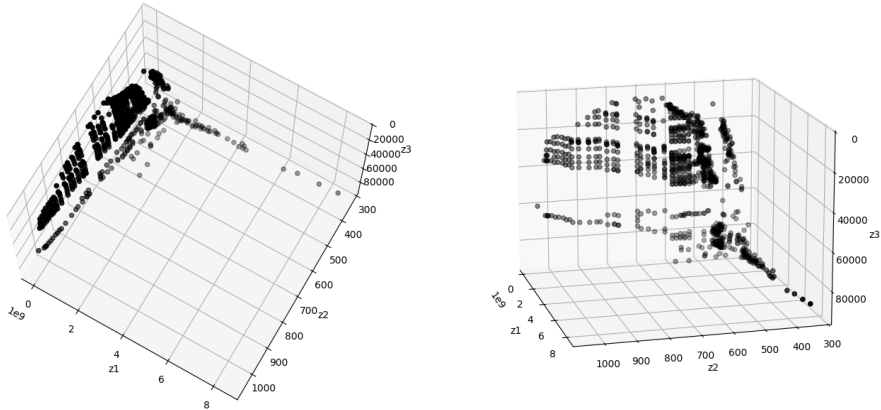


Figure: Pareto set approximated by NSMA for Incomplete variant,  $n = 54$

## Trade-off solutions found by the algorithm

| ID | $Z_1$              | $Z_2$              | $Z_3$              | Opened hubs                |
|----|--------------------|--------------------|--------------------|----------------------------|
| 1  | $6.26 \times 10^7$ | $1.01 \times 10^3$ | $2.55 \times 10^4$ | 15                         |
| 2  | $1.84 \times 10^8$ | $9.07 \times 10^2$ | $4 \times 10^4$    | 12 23                      |
| 3  | $4.62 \times 10^8$ | $6.59 \times 10^2$ | $7.67 \times 10^4$ | 11 25 33                   |
| 4  | $7.35 \times 10^8$ | $4.85 \times 10^2$ | $8.34 \times 10^4$ | 11 25 33 43                |
| 5  | $1.34 \times 10^9$ | $6.07 \times 10^2$ | $9.05 \times 10^4$ | 11 24 27 35 44 48 52       |
| 6  | $2.76 \times 10^9$ | $3.60 \times 10^2$ | $9.05 \times 10^4$ | 10 12 14 27 35 45 47 52 53 |

Table: Some Pareto solutions found by NSMA for Incomplete variant

► As more costs being invested to open hub facilities and building hub network ( $Z_1$ ), the service quality ( $Z_2$ ) and customer's willingness to use the service ( $Z_3$ ) increases. Decision makers can make choices based on their preferences.

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# Conclusion

- This work has reviewed some of the recent development trends in the field.
- By developing a comprehensive mathematical formulation with practical considerations, we obtain a highly useful and realistic model. This model well reflects the real-world problem of railway infrastructure planning.
- Next, we are going to integrate hub capacity constraints into the model. There are multiple levels of hub capacities (small, medium, large). They are decision variables in the model
- Target for a Q2 journal article in the next 1-2 months

# Bibliography I

- [1] Sibel A Alumur, Bahar Y Kara, and Oya E Karasan. “The design of single allocation incomplete hub networks”. In: *Transportation Research Part B: Methodological* 43.10 (2009), pp. 936–951.
- [2] Sibel A Alumur et al. “Perspectives on modeling hub location problems”. In: *European Journal of Operational Research* 291.1 (2021), pp. 1–17.
- [3] James F. Campbell. “Integer programming formulations of discrete hub location problems”. In: *European Journal of Operational Research* 72.2 (1994), pp. 387–405. ISSN: 0377-2217. DOI: [https://doi.org/10.1016/0377-2217\(94\)90318-2](https://doi.org/10.1016/0377-2217(94)90318-2). URL: <https://www.sciencedirect.com/science/article/pii/0377221794903182>.



## Bibliography II

- [4] Kalyanmoy Deb et al. “A fast and elitist multiobjective genetic algorithm: NSGA-II”. In: *IEEE transactions on evolutionary computation* 6.2 (2002), pp. 182–197.
- [5] İbrahim Demir, Fatma Corut Ergin, and Berna Kırız. “A new model for the multi-objective multiple allocation hub network design and routing problem”. In: *IEEE Access* 7 (2019), pp. 90678–90689.
- [6] Robert W Floyd. “Algorithm 97: shortest path”. In: *Communications of the ACM* 5.6 (1962), pp. 345–345.
- [7] Andreia P Guerreiro, Carlos M Fonseca, and Luís Paquete. “The hypervolume indicator: Computational problems and algorithms”. In: *ACM Computing Surveys (CSUR)* 54.6 (2021), pp. 1–42.

## Bibliography III

- [8] Hisao Ishibuchi et al. “Modified distance calculation in generational distance and inverted generational distance”. In: *International conference on evolutionary multi-criterion optimization*. Springer. 2015, pp. 110–125.
- [9] Morton E. O'Kelly. “A quadratic integer program for the location of interacting hub facilities”. In: *European Journal of Operational Research* 32.3 (1987), pp. 393–404. ISSN: 0377-2217. DOI: [https://doi.org/10.1016/S0377-2217\(87\)80007-3](https://doi.org/10.1016/S0377-2217(87)80007-3). URL: <https://www.sciencedirect.com/science/article/pii/S0377221787800073>.

## Bibliography IV

- [10] Chanta Sunarin et al. “A Biobjective Hub Location Model with Consideration of Congestion for Railway Transportation Planning”. In: *Journal of Advances in Information Technology* 15 (Jan. 2024), pp. 414–421. ISSN: 1798-2340. DOI: 10.12720/jait.15.3.414-421. URL: <https://cir.nii.ac.jp/crid/1870021790741118976>.
- [11] Lyndon While et al. “A faster algorithm for calculating hypervolume”. In: *IEEE transactions on evolutionary computation* 10.1 (2006), pp. 29–38.
- [12] Wenying Zhang, Xifu Wang, and Kai Yang. “Uncertain multi-objective optimization for the water–rail–road intermodal transport system with consideration of hub operation process using a memetic algorithm: W. Zhang et al.”. In: *Soft Computing* 24.5 (2020), pp. 3695–3709.